

Julianne Zhu

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EDUCATION

University of Waterloo

Waterloo, ON

BCS, Honours Computer Science with Co-operative Program, GPA: 3.7/4.0

Sep. 2025 – Present

President Scholarship of Distinction

EXPERIENCE

Network Engineering Intern

May 2026 – Sep. 2026

ORBCOMM, SKYWAVE, Network Services Department

Ottawa, ON

- Developed 8 Grafana dashboards with MySQL data pipelines to monitor satellite network performance across 12 metrics, cutting load times by 86% through dynamic filtering and optimized SQL queries
- Shipped an AI-driven analytics service using FastAPI, APScheduler, PyMySQL, Jinja2, and MariaDB that extends predictive forecasting by 60x, cuts pipeline runtime by 78%, and lowers false-positive anomaly alerts by 80%
- Delivered the system architecture and AI-driven monitoring features to engineering leadership, supported by a system design document and live technical demo

Undergraduate Research Mentee

Jan. 2026 – Present

Usable Security & Privacy Research, University of Waterloo

Waterloo, ON

- Co-authoring a research paper for the Symposium On Usable Privacy and Security on how users seek security and privacy advice from Large Language Models (LLMs)
- Presented at the Annual Celebration of Women in Computing 2026 Conference (CAN-CWiC) and IEEE S&P
- Constructed a validated golden-answer dataset from 54 online databases and analyzed 2 LLMs against 100 privacy and security prompts across 15 demographic and thematic categories, using a rubric co-developed with 3 security and HCI researchers to score comprehensibility, actionability, and technical efficacy

PROJECTS

CaneOS | *Swift, Python, C++, SwiftUI, MongoDB Atlas, Auth0, Gemini API, Vercel, Websockets*

Jul. 2026

Full Stack & Embedded Developer

Toronto, ON

- Built a clip-on white cane module to detect overhead hazards within a 4m range for visually impaired users, pairing ToF sensors and 30 FPS YOLOv6 camera detection on an Arduino UNO Q with a Python/FastAPI pipeline to classify hazards via Gemini vision and narrate within 3 seconds via ElevenLabs
- Developed companion SwiftUI iPhone and Apple Watch apps with directional haptics, 4-tier SOS alerts, and Auth0-secured sync of incidents and contacts to MongoDB Atlas via a Node.js serverless proxy

TrulyHer | *React, TypeScript, PostgreSQL, pgvector, Node.js, PyTorch, ONNX, Vercel*

Sep. 2025 – Aug. 2026

AI Integration & Full Stack Developer

Waterloo, ON

- Built full-stack Retrieval-Augmented Generation (RAG) application using React, Node.js, and PostgreSQL with pgvector to support women in CS managing imposter syndrome, enabling semantic search across 200+ journal entries via HNSW vector embeddings and generating personalized reports across 5 analytics views
- Detected imposter syndrome at 0.92 F1 on 200 hand-labeled entries by training a DistilBERT classifier with an object-oriented ML pipeline, calibrating confidence scores and exporting to ONNX with INT8 quantization

Project Tech Careers | *JavaScript, HTML, CSS, VS Code*

May – Jun. 2024

Platform Designer and UI/UX Developer

Ottawa, ON

- Designed a 4-stage platform, curating 100+ resources, to help women in CS find mentorship, scholarships, programs, and community resources across different stages of their education
- Awarded the Gender Equality Track Award over 45 competing teams for innovation, design, and impact in addressing the UN Sustainable Development Goals

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, HTML/CSS, SQL, Bash, Swift, R

Frameworks and APIs: React, Node.js, REST APIs, PyTorch, SQLAlchemy, Gemini API, Uvicorn, Resend API

Data and Analytics: PostgreSQL, MongoDB Atlas, pgvector, InfluxDB, pandas, NumPy, Grafana, scikit-learn

Tools and Systems: Git/GitHub, Linux, AWS, Google Cloud Platform, Vercel, VS Code, Xcode, VirtualBox